Current Status

This repository now has two clearly separated tracks:

- **Task 1:** implemented and locally validated. The active compliant baseline uses only the two official Task 1 checkpoints:
  1D_Burgers_Sols_Nu0.001_FNO.pt and 1D_Burgers_Sols_Nu0.001_Unet-PF-20.pt.
- **Task 2:** scaffolded only. The final package may include a persistence prediction so the official file flow is valid, but this is not a competitive model yet.

Older multi-nu and nu=0.1 fine-tuning experiments are historical exploration records. They should not be treated as active final-submission evidence unless the race rules are clarified to allow those assets.

Task 1 Active Pipeline

Input: data/Task1/task1_test.hdf5 with shape (1000, 10, 256).

Output: task1_pred.hdf5 with shape (1000, 200, 256). The first 10 frames are copied from the input and the remaining 190 frames are forecast.

Current active validation candidates:

```text
final blend:          nu0.001 FNO = 0.12, Unet-PF-20 = 0.88
proxy-score blend:    nu0.001 FNO = 0.04, Unet-PF-20 = 0.96
Unet-only baseline:   nu0.001 FNO = 0.00, Unet-PF-20 = 1.00
```

Local validation on data/Task1/task1_val.hdf5:

Candidate	MSE	Proxy Score
best-MSE blend 0.12/0.88	0.0582305179	13.03828949
proxy-score blend 0.04/0.96	0.0611382528	13.83318215
Unet-only 0.00/1.00	0.0648512424	13.39284111
FNO-only 1.00/0.00	0.4238491430	5.13830407

Task 2 Scaffold

Input: data/Task2/task2_test.h5 with shape (1000, 10, 256).

Output: task2_pred.hdf5 with shape (1000, 200, 256). The official sample submission keeps the first 10 output frames equal to the input, so the current scaffold follows that convention and repeats the last observed frame for the remaining 190 frames.

Implemented scaffold:

- code/task2_persistence_baseline.py
- agent/task2_workflow.py
- agent/task2_submission.py

This is only a correctness baseline for validation and packaging. The competitive Task 2 model still needs a dedicated training strategy using task2_part0_train.h5, task2_part1_train.h5, and task2_part2_train.h5.

Final Packaging Flow

The official packaging path is:

powershell
D:\Junao\ProgramData\anaconda3\envs\Hwpytorch\python.exe -m agent.final_submission --run-name final-official-ensemble-task2-persistence --task1-weights 0.12 0.88 --task2 persistence

This creates:

text
```
runs/final-official-ensemble-task2-persistence/
  submission.json
  task1_pred.hdf5
  task1_time.csv
  task1_logs.log
  task2_pred.hdf5
  task2_time.csv
  task2_logs.log
  methodology.pdf
  code/
  pred.zip
```

The final validator is:

powershell

```
D:\Junao\ProgramData\anaconda3\envs\Hwpytorch\python.exe -m agent.validate_submission --path
runs\final-official-ensemble-task2-persistence
```

Loss Design For New Training

agent/task1_baseline_train.py now supports optional loss terms:

- supervised trajectory MSE over a chosen time window
- initial-condition consistency loss
- spectral MSE with extra high-frequency weight
- Burgers residual loss u_t + u u_x - nu u_xx

Example:

powershell
```
D:\Junao\ProgramData\anaconda3\envs\Hwpytorch\python.exe -m agent.run_task1_baseline_zoo
--study-name task1-physics-loss-v1 --models residual_refiner --max-samples 4096 --steps 2000
--batch-size 16 --device cuda --loss-start-step 10 --loss-end-step 200 --physics-loss-weight
0.001 --spectral-loss-weight 0.01 --spectral-high-weight 4.0 --base-train-hdf5
runs\base_cache\task1_train_base.hdf5 --base-validation-prediction-path
runs\_verify_official_ensemble\task1_val_fno_unet_012_088.hdf5
```

Reproducibility Commands

Validate the active Task 1 MCTS path:

powershell
```
D:\Junao\ProgramData\anaconda3\envs\Hwpytorch\python.exe -m agent.run_task1_mcts_experiment
--config configs\task1_mcts_validation_smoke.yaml --reset
```

Run direct Task 1 official checkpoint inference:

powershell
```
D:\Junao\ProgramData\anaconda3\envs\Hwpytorch\python.exe code\official_checkpoint_ensemble.py
--input data\Task1\task1_test.hdf5 --output runs\task1-official-ensemble\task1_pred.hdf5
--batch-size 64 --models fno=checkpoints\extracted\1D_Burgers_Sols_Nu0.001_FNO.pt
unet_pf20=checkpoints\extracted\1D_Burgers_Sols_Nu0.001_Unet-PF-20.pt --weights 0.12 0.88
```

Validate Task 2 scaffold:

powershell

```
D:\Junao\ProgramData\anaconda3\envs\Hwpytorch\python.exe code\task2_persistence_baseline.py
--input data\Task2\task2_test.h5 --output runs\task2-persistence\task2_pred.hdf5
```